Ollscoil na Gaillimhe
University of Galway

Power Electronics and Closed Loop Current Control for the Galway Energy Efficient Car

Conor Farrell - 21359763

Supervisor – Prof. Maeve Duffy

Co-Assessor– Prof. Edward Jones

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theGeec

1 Abstract

This thesis details the design, implementation, and testing of a new power electronics system and closed-loop current controller for the Galway Energy Efficient Car (Geec), a student-built solar electric vehicle competing in the Shell Eco Marathon. The system centres on a synchronous buck converter implemented across two custom PCBs: a power electronics board and a power controller board, the latter built around an Arduino Nano ESP32.

A motor selection study was conducted comparing the existing brushed Maxon RE50 and the brushless Maxon EC DT 85 Frameless, using race data from the 2024 Nogaro event to back-calculate real-world energy consumption. The brushed motor was retained for continuity, though recommendations for a future brushless transition are provided.

MOSFET selection was evaluated across four candidates including a GaNFET, with the ISG0613N04NM6H integrated half-bridge selected based on switching loss analysis, solder compatibility, and voltage headroom. Cycle-by-cycle peak current limiting was designed and implemented in hardware using a DAC-configurable threshold, a latching comparator, and a D-type flip-flop synchronised to the PWM signal, addressing the high current spikes of 30–40A observed under open-loop acceleration.

Dynamometer testing demonstrated a 12% improvement in energy consumption compared to the previous open-loop 5-second ramp rate under equivalent constant-torque conditions, with further gains expected under dynamic race conditions. Idle power consumption of the power electronics board was reduced from 0.91W to 0.225W. Several hardware issues were encountered and resolved during bring-up, including incorrectly oriented TVS diodes, an op-amp pin error, a cold solder joint on the DAC, and a logic level mismatch between the 3.3V ESP32 and 5V flip-flop.

The modular separation of the power electronics and power controller boards established by this project provides a foundation for independent development of each subsystem in future iterations.

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3 Introduction

3.1 The Geec & SEM

The Geec is the Galway energy efficient car, it has been a multi denomination team that has been active in the university since 2014. This project typically has between 4 and 7 final year projects divided between mechanical and electrical theses.

The car competes at the Shell Eco Marathon. This year the competition will be hosted at the Silesia Ring in Poland. This is a very flat track with a total elevation difference of 2m over a 700m lap.

3.2 Power Electronics

The power electronics system is responsible for controlling the energy delivery from the battery to the drive motor in the Geec solar electric vehicle. At its core, the system is a synchronous buck converter, converting the battery's supply voltage to a controllable output via pulse-width modulation (PWM). The design encompasses two custom PCBs: a power electronics board housing the switching MOSFETs and passive components, and a power controller board integrating an ESP32 microcontroller, current sensing circuitry, and isolated communication interfaces.

A key objective of this iteration was improving on the existing power electronics in three areas: switching efficiency through better MOSFET selection, power consumption at idle, and closed-loop current control to reduce energy waste during acceleration. The previous system lacked current limiting, relying instead on an open-loop duty cycle ramp, which led to high current spikes of 30–40A during acceleration and inefficient energy use on track.

Another goal of this project is to separate the development of the power control system to the development of the power electronics system. This can separate the project further and allow for new projects to investigate practical improvements in two directions without impacting on the other thesis active work.

3.3 Previous Projects

The two primary theses that were relevant to this thesis were Rory Egan's thesis from 2025, this provided simulation results suggesting that there were significant gains to be made by transitioning to brushless motors. Through my own investigations I noticed a mistake in the calculations used to determine the power losses in the brushed motor, comparing with Maxon's recommendation for simulation of power losses. The results suggested the motor was less efficient than the motor has been found on track.

Oisin Anderson designed the previous power electronics which the new circuit was designed and benchmarked against. The goals of his project were

3.4 Results

The new power electronics achieved significant improvements over the previous generation across all three target areas:

Current Control – Cycle-by-cycle peak current limiting was successfully implemented and validated on the dynamometer. Compared to the previous open-loop 5-second ramp, the current-limited system demonstrated a 12% improvement in energy consumption under equivalent constant-torque load conditions. Under dynamic conditions representative of actual racetrack use, further improvements are expected.

Switching Frequency – Testing across multiple switching frequencies confirmed the expected trade-off between ripple current and switching losses. At 10kHz the ripple current corresponded to approximately 11W dissipation at the motor terminals, compared to 8.5W at 45kHz, consistent with theoretical predictions.

Debugging & Validation – Several hardware issues were identified and resolved during bring-up, including incorrectly oriented TVS diodes, an op-amp supply pin error on the power controller, a cold solder joint on the DAC, and a logic level incompatibility between the 3.3V ESP32 and the 5V flip-flop. All issues were resolved and the system was validated through both bench and dynamometer testing.

4 Design

4.1 Motor Selection

Following on from Rory Egan's thesis last year, an investigation into the efficiency of a brushed and brushless motor was conducted. The motors selected for comparison were the existing brushed Maxon RE50 (370354) and the brushless Maxon EC DT 85 Frameless (740953) shown in Figure 4.1-1. After evaluating the datasheets and comparing across against 15 other brushless motors and several other brushed motors, these two were identified as the most promising standout options.

Figure 4.1-1 Two motors evaluated as optimal brushed and brushless motors

A detailed evaluation of the two motors was conducted, with the graph in figure 4.1-2 showing the comparative efficiency of both motors. The brushed Maxon RE50 (370354) demonstrated a higher maximum efficiency, while the brushless Maxon EC DT 85 Frameless (740953) showed better acceleration efficiency up to 45% duty cycle. The actual differences between the two motors were relatively minimal, with only 0.5% higher efficiency at 88% duty cycle.

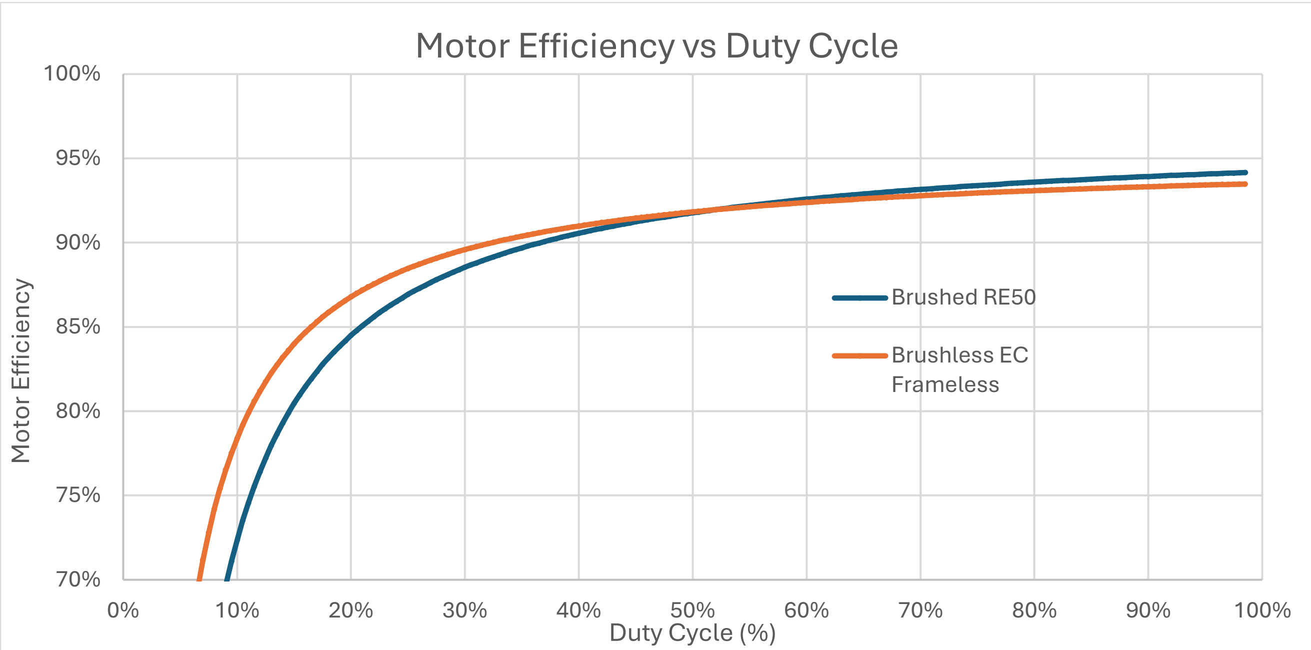


Figure 4.1-2 Efficiency curve of brushless and brushed motors.

Using race data from Nogaro (2024), where the existing motor setup was used, it was possible to back-calculate and estimate the energy consumption of the alternative motor under the same conditions. The data was first used to determine the torque and angular speed of the wheel, which then allowed an optimal gear ratio to be established. From these values, the required input voltage and current needed to achieve the same performance as the existing setup were calculated. The analysis showed that the alternative motor would have consumed approximately 1% more energy overall.

An engineering decision had to be made on whether to continue with the existing motor or to swap to the brushless. Table 4.1-1 details the pros and cons of each motor

Existing Brushed Motor (RE-50 370354)	
Pros	Cons
• **More Efficient (1%)** • **Easier to Control** • **Existing Data on Motor**	• High motor speed (5680 vs 2850 rpm) • High current • Not Recommended for New Design (Issue found in May 2026)

New Brushless Motor (EC-DT85 740953)	
Pros	Cons
• **Various Control Strategies** • **Highly integrated ICs from TI, ST and ADI for FOC and Trap** • **Cheaper (€380 vs €632)**	• Less efficient • New Drivetrain Required • New batteries (48V system)

Table 4.1-1 Table comparing pros and cons of brushed and brushless motors.

The primary factors concerning the decision were; 1) The lack of data on a new motor and the lack of use of existing data, and 2) There was not an existing drivetrain project on the mechanical department meaning the motor would not be implemented on the car this year. The motor hadn't been placed into "Not Recommended for New Design" at the time of this decision in November 2025. Knowing this fact now, I might've further advocated for the brushless option.

If a pivot to brushless is selected next year my suggestions as a starting point are. All options are rated up to 60V:

- TLE9180D: Infineon, integrated DCDC buck converter, smart deadtime, integrated CSA - $4.20 on JLCPCB, 6EDL7141.
- DRV8334: TI, like TLE9180 no buck converter, Can control over SPI - $2.40.
- TMC6200: ADI, Supports inline current sensing, LDO regulator, - $6.70.

For all these solutions, they directly support trapezoidal control with hall effect sensors, that can be added to the EC DT85, for field orientated control this can be done with the simplefoc library for Arduino, ESP32 and STM32. External MOSFETs are required which increases cost but generally have lower Rds_{on} than the BLDC drivers with integrated FETs.

4.2 Parts Selection

4.2.1 MOSFETs

In 2021, Óisín Anderson's power electronics PCB used D^2PAK Infineon IPB014N08NM6 MOSFETs, which featured low Rds_{on} of 1.4mΩ, although the gate capacitance (Q_G: 151nC) is very high and the package is very large. In 2023 David Kong replaced these with smaller footprint PSMN1R0-30YLD MOSFETs, which gave a better performance on paper. However, due to the mirrored gate pin arrangement and the incompatibility between the new MOSFETs and the PCB footprint, the component was not soldered correctly. As shown in Figure 4.2-3, the solder joints do not touch the bulk of the heat spreader, increasing electrical resistance. Consequently, the MOSFET in operation was likely higher than the 1mΩ specified in the datasheet.

In addition, the PSMN1R0-30YLD MOSFETs were rated for a maximum voltage of 30V, while the system voltage could reach up to 29.4V . This provided almost no headroom in the event of possible voltage spikes generated during operation, putting the MOSFET out of tolerance of its specified voltage limit.

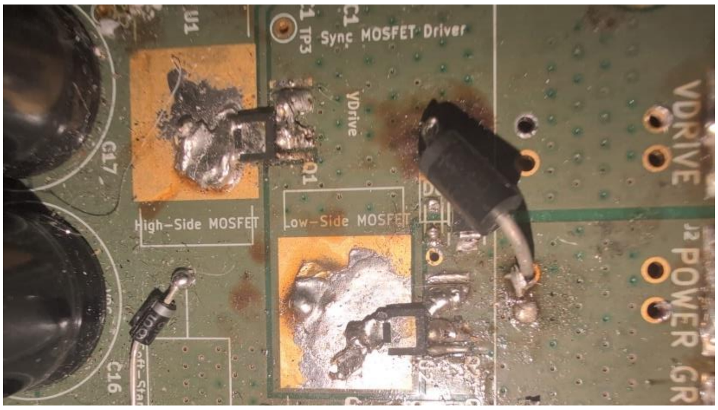


Figure 4.2-1 New MOSFETs on old power electronics.

Three MOSFETs and one GaNFET were investigated. IPB014N06N(D^2PAK), PSMN1R0-30YLD(LFPAK56), ISG0613N04NM6H and the GaNFET EPC2367. The GaNFET has the benefit of very low gate capacitance and no body diode, but a higher Rds_{on}. Using [this white paper] from Texas Instruments, the results in figure 4-4 were calculated, for a supply voltage of 29.4V and an output current of 20A with a ripple current of:

$$\Delta I = \frac{V_{CC}}{4.L_{TOT}.F_{SW}}$$

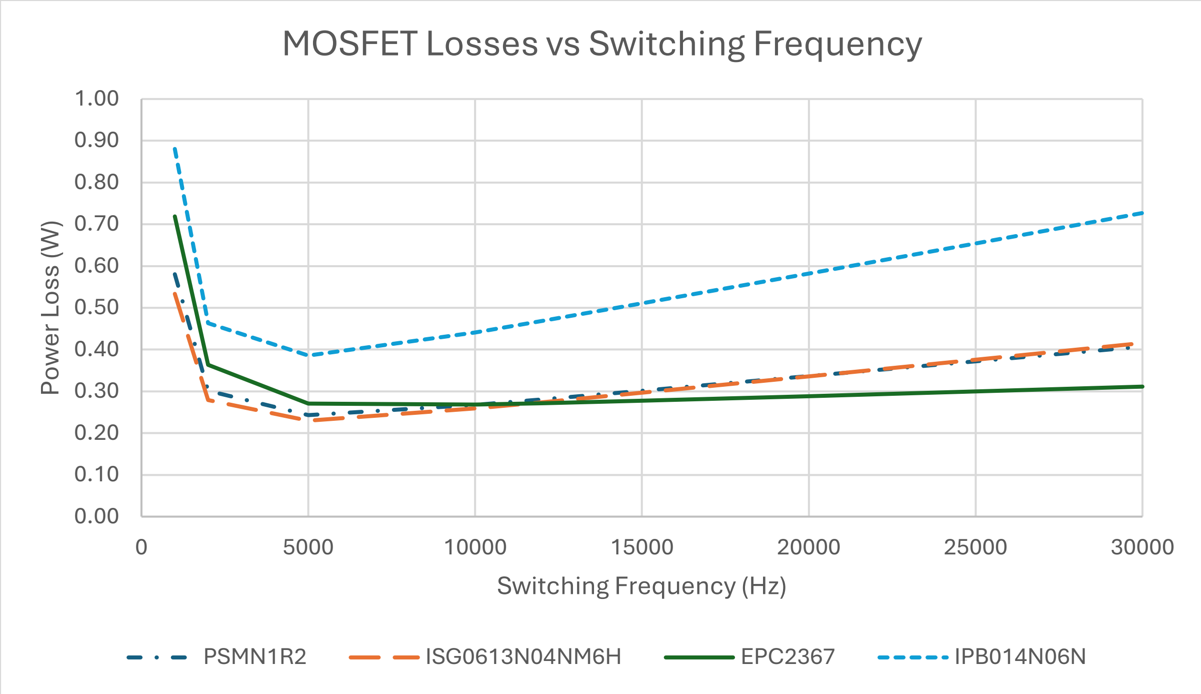


Figure 4.2-2 Graph of MOSFET power loss versus switching frequency.

The best FET for the frequency that the motor is currently switching at is the EPC2367, with a very low increase in power loss as the switching frequency increases, primarily because of the lack of a body diode. The next best FET is the ISG0613N04NM6H which is slightly better than the PSMN1R2. This MOSFET is also an integrated half bridge.

The ISG0613N04NM6H was chosen to move forward with due to the standard MOSFET switching behaviour compared to the tighter requirements of a GaNFET, requiring a Vgs_{thres} of 5 and a Vgs_{max} of just 6V, additionally the packages for GaNFETs are difficult to hand solder and verify in SPICE. For these reasons the ISG0613N04NM6H was chosen, with the option to solder PSMN1R2 or other MOSFETs with the same footprint.

In the case of further investigation into GaN being of interest in further years the STDRIVEG600 is an ideal gate driver in a SOIC-16 package which is useful for hand soldering. An alternative GaNFET is the Infineon IGC019S06S1.

4.2.2 Isolation

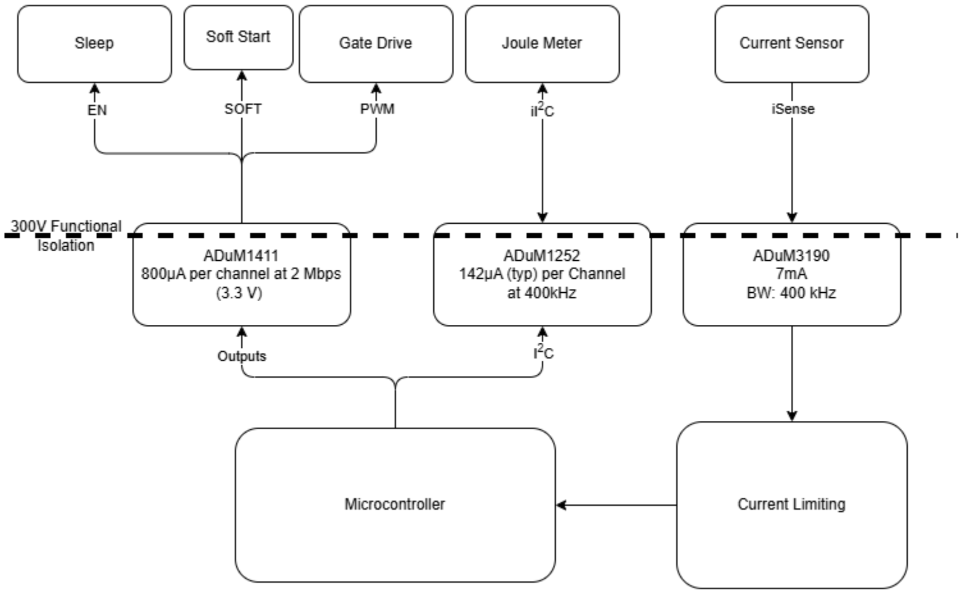


Figure 4.2-3 Isolation Input and Output Diagram.

The isolation layer is critical to ensure safety between the battery and motor and the microcontroller. The ADuM1411 is a 3 input 1 output isolator IC. It's primarily designed for SPI but for the purpose of the power electronics the PCB it works perfectly. The IC also sets the output to low by default, this prevents initial power up from causing the motor to enable on startup.

The ADuM1252 is a low power I^2C isolator, this can connect directly to the joulemeter and provide live updates. The ADuM3190 is a purpose-built analogue isolator Op amp. This is one of the few ICs that have LTSpice models and also output a single ended voltage.

4.3 Current Control

4.3.1 Current Sensors

On the current power electronics and DAQ the current sensors are the ACS770 and ACS723, these are hall effect sensors, which have full isolation between the inputs and output. The sensors only have 120 or 80kHz bandwidth. The current waveform is relatively triangular, this means only the only the first and third harmonic are within the sensor's bandwidth. In comparison a shunt resistor instrumentation amplifier supports bandwidths in the MHz range and output voltages based on a fixed resistor. The INA241B3 has a gain of 50V/V and a bandwidth of 1.1Mhz, the AD8411A has a 2.7Mhz bandwidth and a gain of 50V/V.

Neither sensor is isolated nor requires an additional sensor to convert to the isolated reference, this is difficult problem with most sensors providing a differential isolation output. The ADuM4190 is an isolation amplifier which provides a single ended output with a gain of ~2.5. The bandwidth of the ADuM4190 is 400kHz giving a response up to the 13^{th} harmonic. Figure 4.3-1 shows the triangular wave, which is the expected waveform of an RL circuit under a square waveform as shown by the buck converter whitepaper.

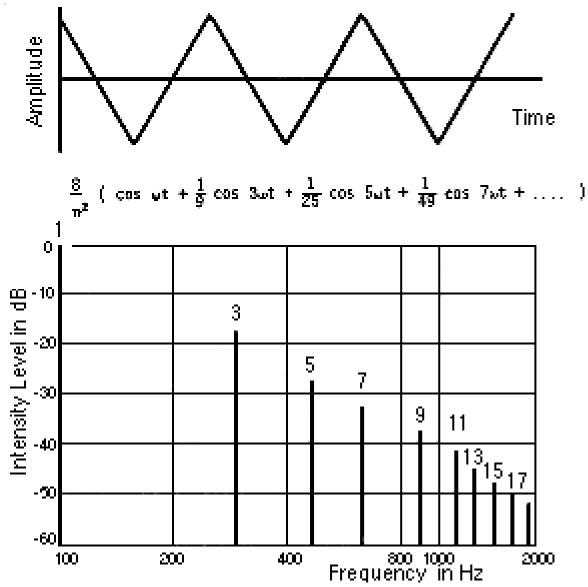


Figure 4.3-1 Harmonics of a triangular waveform. (100Hz)

4.3.2 Cycle By Cycle Limiting

The current drawn by the motor is directly proportional to the torque acting on the motor output. The important motor equations are:

$$\Omega_{\text{rpm}} = K_s.V_{bar}.d - \frac{\Delta n}{\Delta M}.T$$

$$T = K_T.I_{mot}$$

The Geec has a large inertia, this means that the motor is primarily behaving in this datasheet characteristic. In an open loop setup this results in a torque acting on the motor increasing if the speed voltage is higher than the rear wheel is rotating at. In real world results this manifests as high current spikes in accelerations (30-40A), this is intense on the motor, battery and power electronics but also not the most efficient solution.

Acceleration in the Geec used a maximum duty cycle ramp rate of 5 seconds from 0-100%. More aggressive and more conservative accelerations were investigated, but they caused the driver to unintentionally over-accelerate and trigger the BMS Over Current Protection. The issue is primarily that the torque requirement for an acceleration is a dynamic property based on weight, slope, instantaneous speed, friction etc. The open loop system innately struggles with this, an issue that couldn't be diagnosed on the dynamometer due to the constant nature of tests performed on the dyno.

Current limiting leads on from this issue, and with those two methods for limiting: cycle-by-cycle and average current. Cycle by cycle is the method chosen for investigation and implementation in this project, but the project *can* support average current limiting and dynamic current limits with some firmware modifications.

Peak current cycle-by-cycle limiting involves monitoring the current against a threshold and then disabling the system until the next cycle. In the context of the Geec buck converter this means that the high side MOSFET is disabled and the low side FET is enabled. The naïve implementation of this is a comparator connected to an AND gate with the PWM. The primary issues with this circuit are latching and hysteresis.

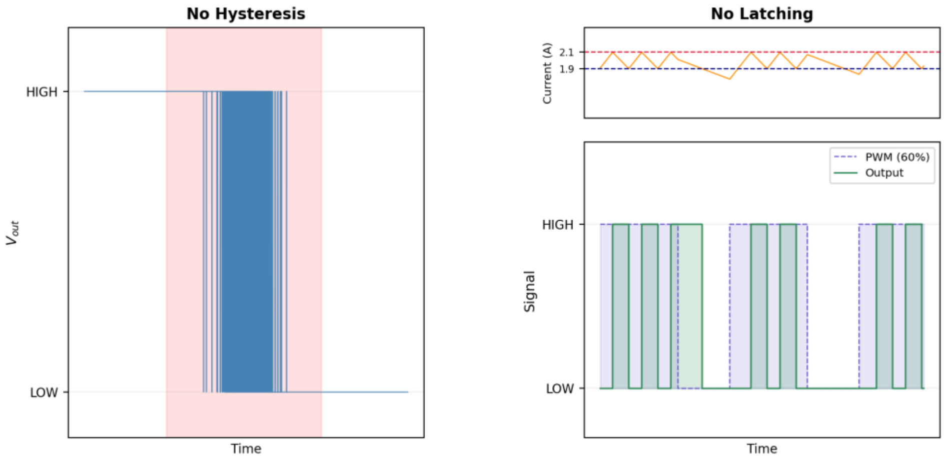


Figure 4.3-2 Comparator output without hysteresis or latching.

Figure 4.3-2 shows the two issues with this naïve solution the first is that the output can rapidly oscillate around the transition voltage, causing high current draw to the IC and unstable results on the output PWM. In the case of the right graph, there is some hysteresis, but another issue arises. As the current decreases the comparator returns to the reset position allowing the output to go high again. This results in a higher effective switching frequency of the circuit, increasing energy usage although flattening the current curve. This effect is difficult to control and changes with the applied duty cycle, current limit and load.

Figure 4.3-3 shows the final schematic for the cycle-by-cycle peak current mode limiting. The first component on the Power controller board is a DAC, this allows for any voltage between 0V and 4.096V to be set as the current limit. The comparator is configured with a V_{th}:0.8V and V_{tl}:0.6V, the negative input is then connected to the output of a differential amplifier. The differential amplifier leverages the supply voltage to limit the negative response to a floor of 0V, when Isense is greater than Vdac there is a gain of 6.8. A D-type flip flop is then used to latch the output, the input D is pulled to Vcc (5V) with the reset pin connected to the output of the comparator. A 1µs pulse is synced with the output PWM that resets the flip flop. This is the minimum response time for output power primarily due to the response rate of the AD8411a.

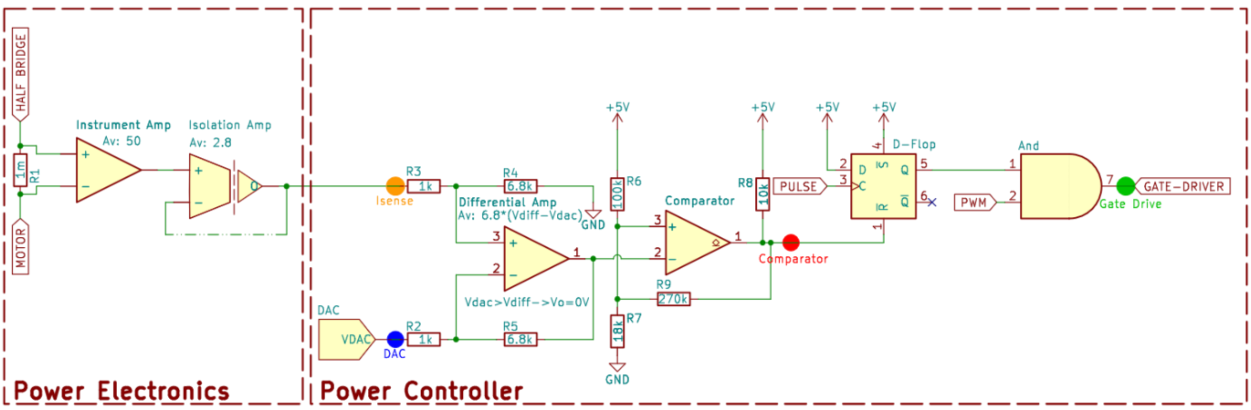
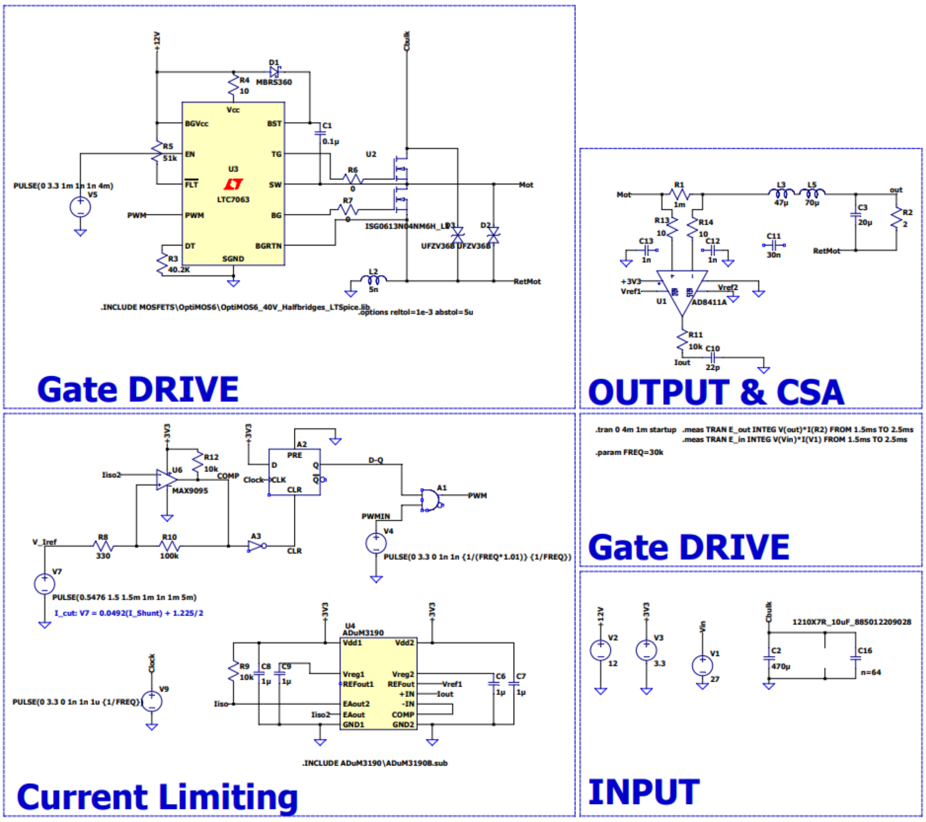
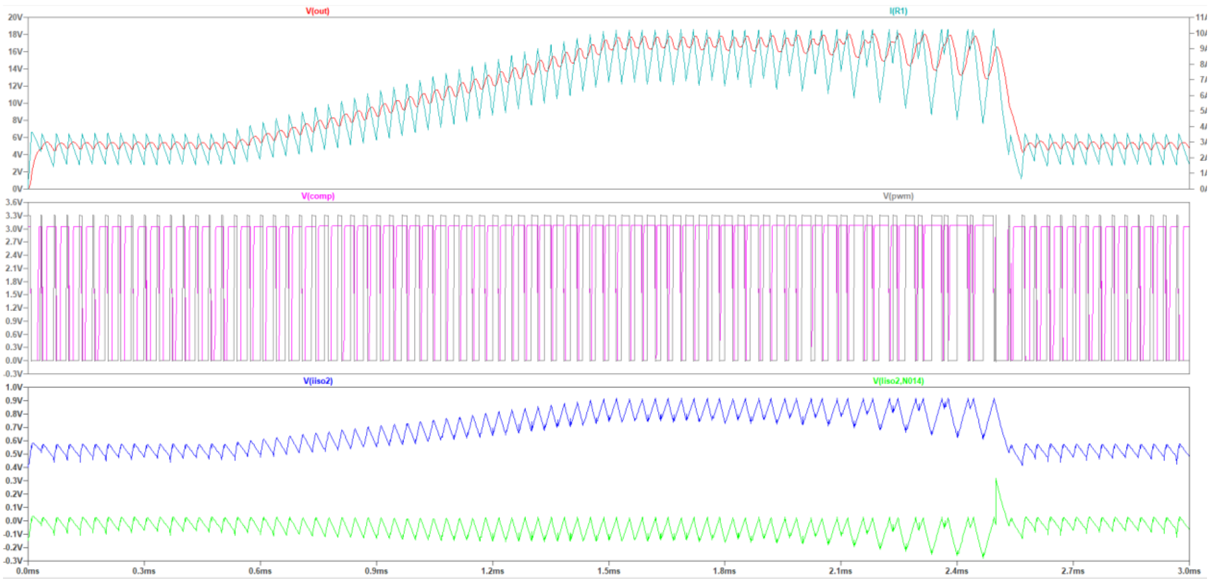


Figure 4.3-3 Final Circuit Implementation of Cycle-by-cycle limiting

4.3.3 Simulation





4.4 PCB Design

The PCBs were designed in Altium, using custom schematic symbols and manufacturer footprints where possible. Two PCB projects were made. One for the power electronics board and another for the power controller board.

4.4.1 Power Electronics

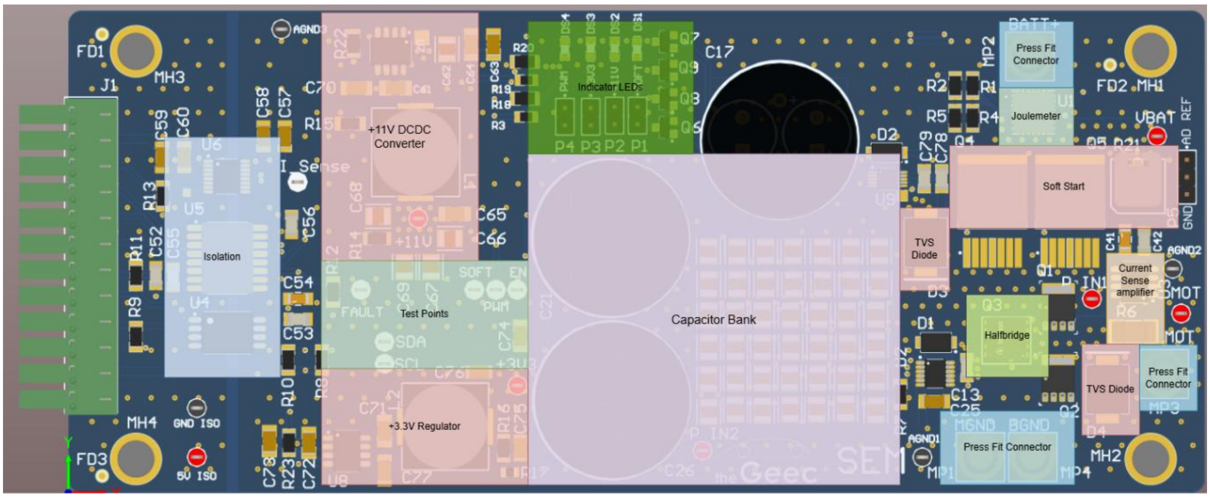


Figure 4.4-1 Functional areas for each part of the PCB

The layout of the power electronics focused on keeping the functional blocks close to each other. Each primary element is clustered allowing for short routing. To route across the board a simple strategy of vertical traces on the top layer and horizontal traces on the bottom plane.

The PCB is a 4-layer board with the 2nd layer containing a solid ground plane and the 3rd layer containing power planes, An 11V power plane at the top and a 3.3V power plane on the bottom. In the middle there's the battery power plane for additional routing between the capacitor and the switch node. For isolation there's a separation between the planes and signals coming from the power controller board and the all signals and supplies on the power board.

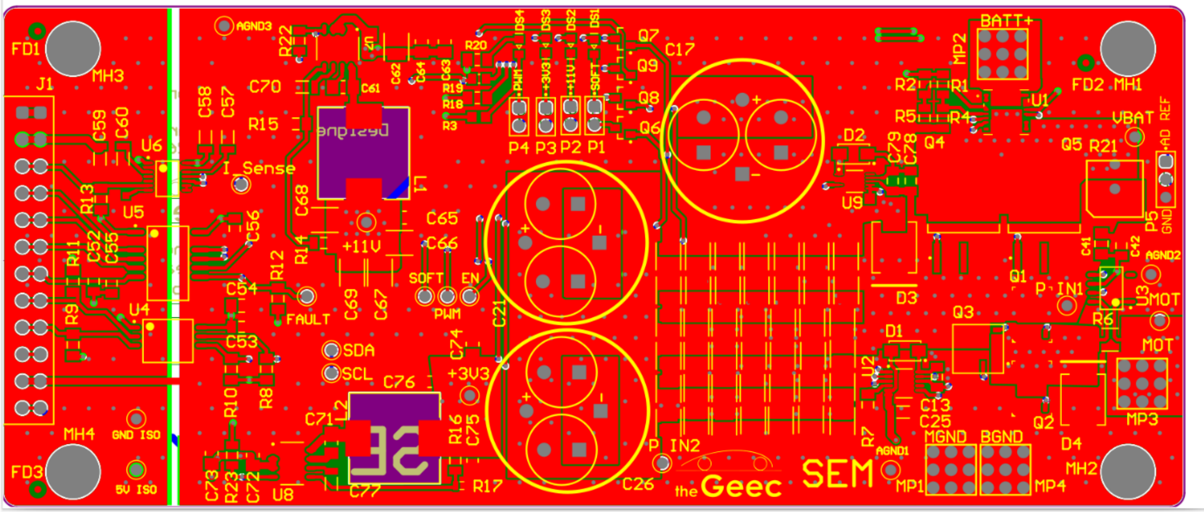


Figure 4.4-2 View of top plane on the power electronics PCB.

4.4.2 Power Controller

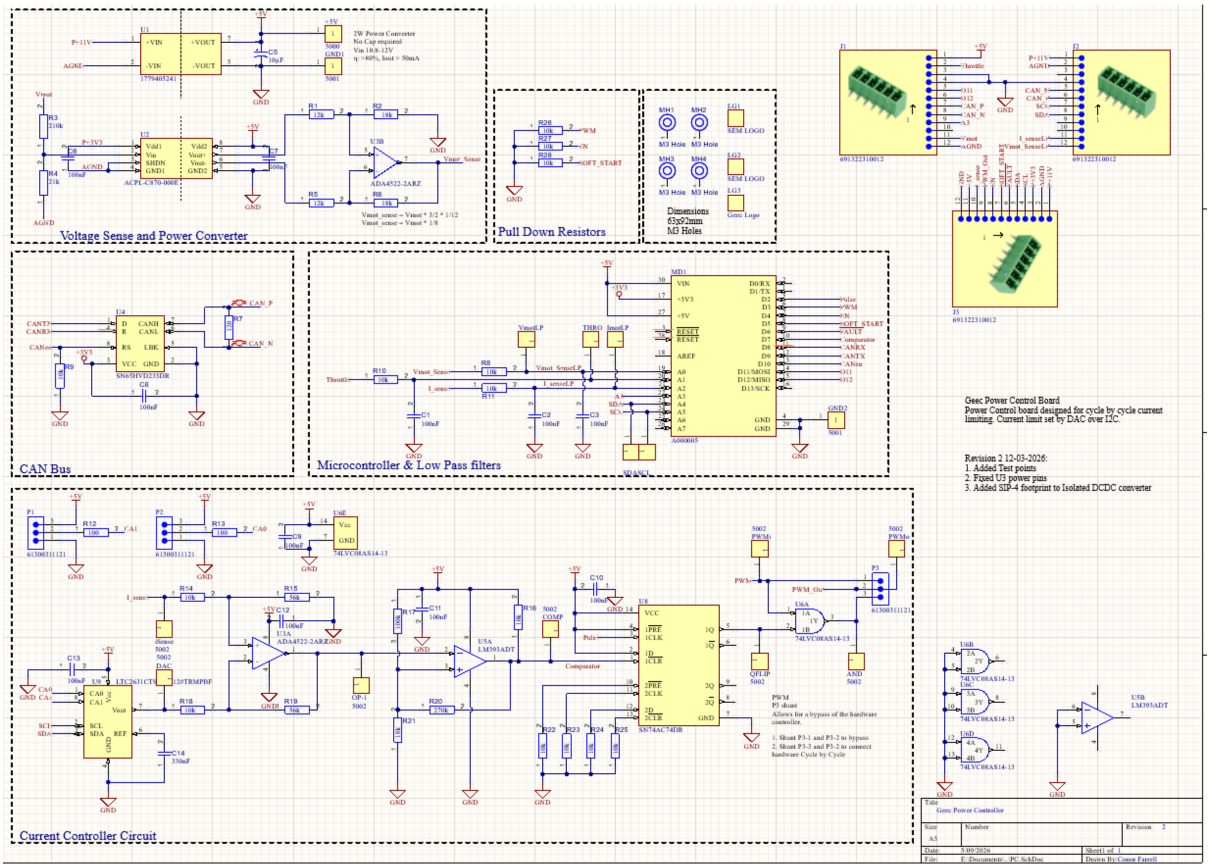


Figure 4.4-3 Schematic of power controller PCB.

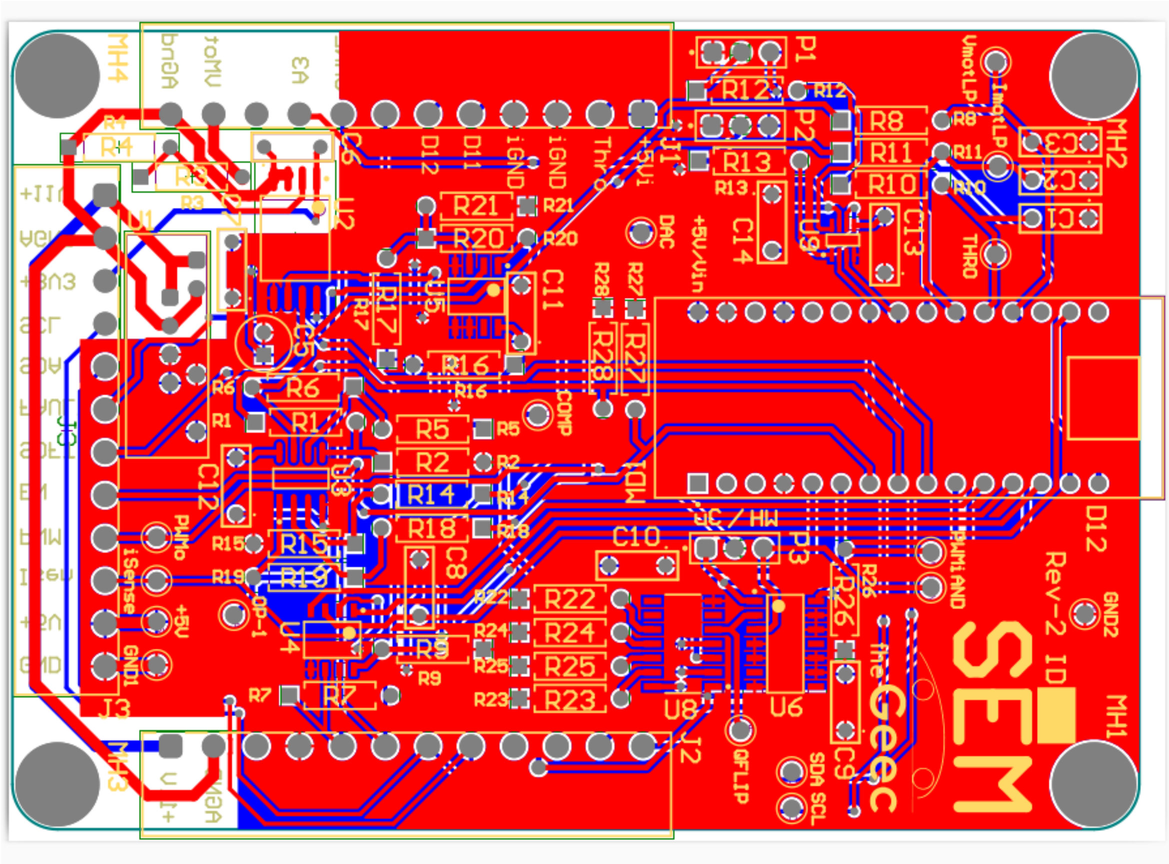
The power controller is a 2-layer PCB. The PCB uses an Arduino Nano ESP32 integrating the current control mechanism detailed in section 4.3. This design is fairly dense, and a lack of a solid ground plane shows some noise in

Other Features

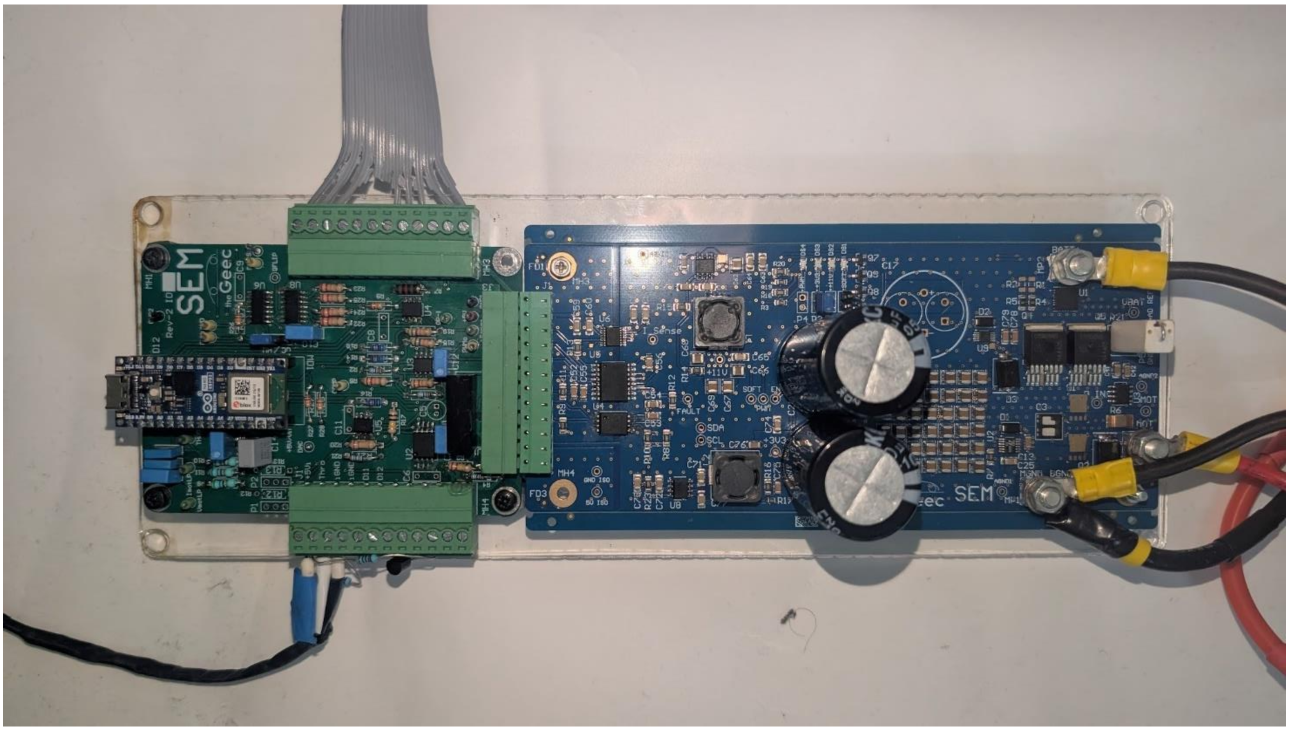
- **CAN:** The SN65HVD233DR is a CAN transceiver, that can be interfaced with using the ESP32 MCPWM peripheral. This allows for native communication over a differential signal which is implemented in the DAC and steering wheel display.
- **Motor Voltage**: The motor voltage is measured from separate connections to the power electronics board. This allows the motor voltage to be measured directly at the terminals improving accuracy due to not including the voltage drop in the cables.
- **Spare Pins:** The Arduino Nano Esp32 has multiple spare analogue and digital pins which are not required for communication or control, some of these have been exposed on header connectors.

The inputs on all disconnected logic gates have been tied to ground. This keeps the gates in a safe logic state and prevents outputs changing, which I have seen cause power issues on breadboard testing.

The 3 12 pin 3.81mm connectors are designed to connect to pluggable terminal blocks. In the case of the front one this is soldered directly to on the PE PCB, a wire-to-wire connector can be used for the left and right connectors. This additionally improves the flexibility of the PCB.



If I was to design this PCB again. I would change the pinout to allow compatibility with both the ESP32 Nano and Nano R4, move everything to 3.3V logic after the comparator not 5V, create a solid ground plane on both the top and bottom layers, or use a 4 layer design. There is some noise in the ground plane that maybe from the DCDC converter.



4.5 Code

4.5.1 Firmware

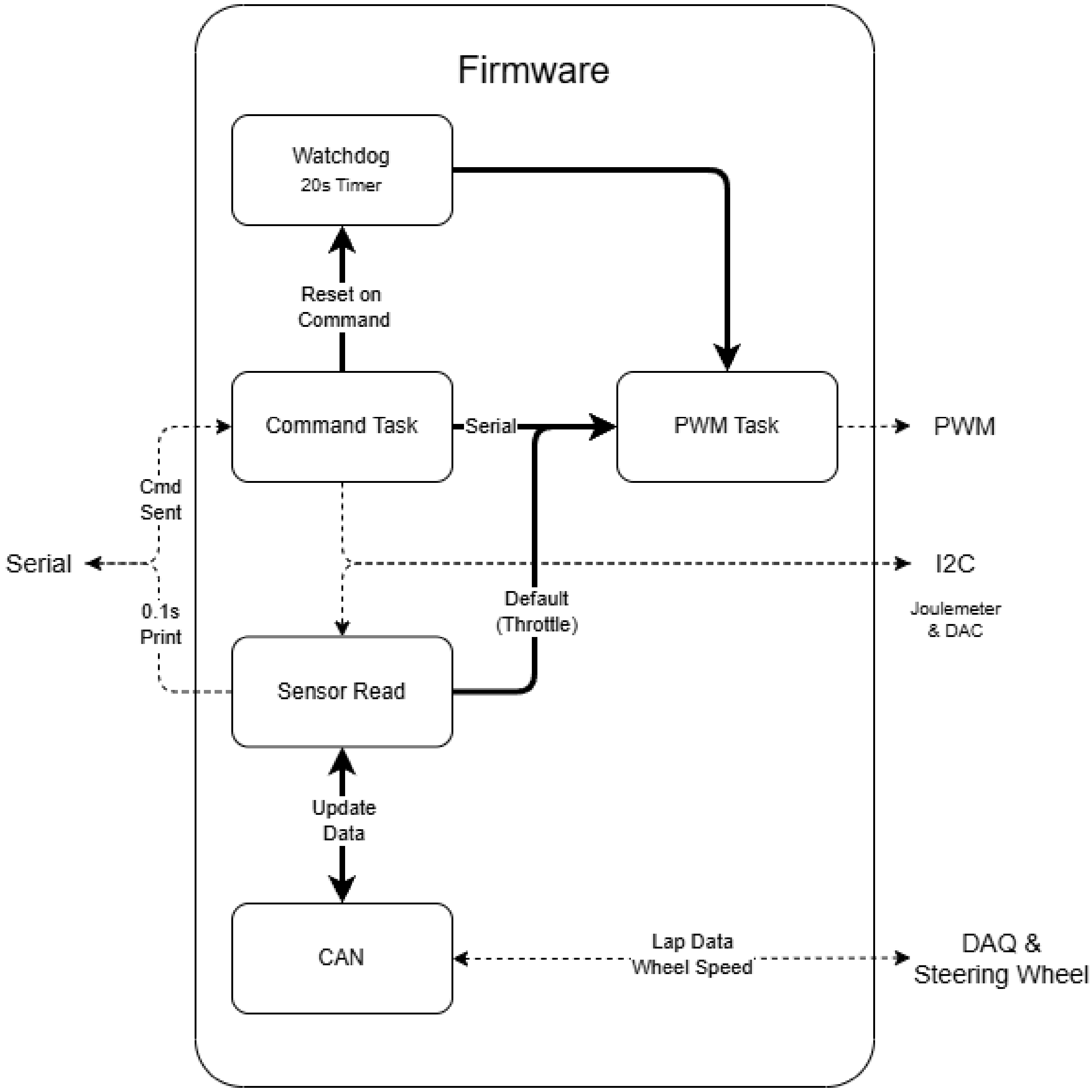


Figure 4.5-1 Flow diagram of the sensor tasks in firmware.

The firmware running on the ESP32 is divided into multiple RTOS tasks, this was primarily done for ease of development as it allowed code to be added sequentially without interacting with other elements.

- **Command Task:** The command task receives messages over serial which can be used to set variables. The commands are:
  - **H:** Help, prints out the list of commands
  - **C <Amps>:** Sets the current limit on the power controller. *Note this is max current not average current. Float.*
  - **M:** Sets the microcontroller to serial controller overwriting the input from the throttle.
  - **0-100:** Sets the target duty cycle, for 30kHz the actual limit is 99.8% duty to bootstrap capacitor charging. *Not enabled on boot.*

- **T:** Sets the microcontroller to throttle control. Disabling Serial commands.
  - **S:** Immediately sets the PWM to 0%.
  - **R:** Resets the joulemeter back to 0J.
- **Sensor Read:** Reads all the important information from the joulemeter, checks the most recent updates to the PWM, wheel speed etc. and prints them to the serial monitor. Runs every 100ms.
- **CAN:** Sends all the important messages every 100ms for processing by the DAQ or steering wheel, checks incoming messages and updates local values like wheel speed. Header file containing message IDs is used.
- **PWM Task:** Reads the current target PWM and increases the actual PWM by the required amount per tick until the output PWM Is equal. Due to the current limiting this ramp rate is 1-2s.
- **Watchdog Task:** For serial testing if the car is in an active state for longer than 20s without receiving a new message the watchdog will trigger and set targetdutycycle to 0. Turning off the motor.

In future the code could be simplified into a single super loop with deterministic timing. However, the flexibility allowed by the RTOS was helpful. Another area to move forward with would be engaging the sleep mode on the ESP32. This microcontroller uses most of the idle power consumed between the power controller and power electronics board.

4.5.2 Test Code

The test code for the power electronics integrated both the power electronics and the dynamometer. The code uses a pyQT GUI to display current, voltage, output power, input power, dyno torque and wheel speed. The code then runs an experiment: with 3 variables to set current limit, minimum speed, maximum speed. The experiment then runs 5 accelerations and decelerations. The code combines the serial output of both the dyno and power controller, saving both sets of results to a csv file. Speed can be displayed and used for calculations by the dyno speed or the DAQ speed.

The primary issue is both sets of speeds only updates once per second

5 Results

5.1 Circuit Debugging

5.1.1 TVS Diodes

When the main power electronics boards were received, they immediately went to a constant current 1V. After debugging with a thermal camera, desoldering a component that got hot. I tried to desolder the TVS diodes and repeat the issue. This resolved the problem. Investigating again I found the problem was that during assembly I was asked to verify the orientation of the TVS diodes. I mistakenly asked for D3 and D4 to be rotated 180°, to have the + sitting on the high voltage side. TVS diode should be reverse biased and the orientation below was correct.

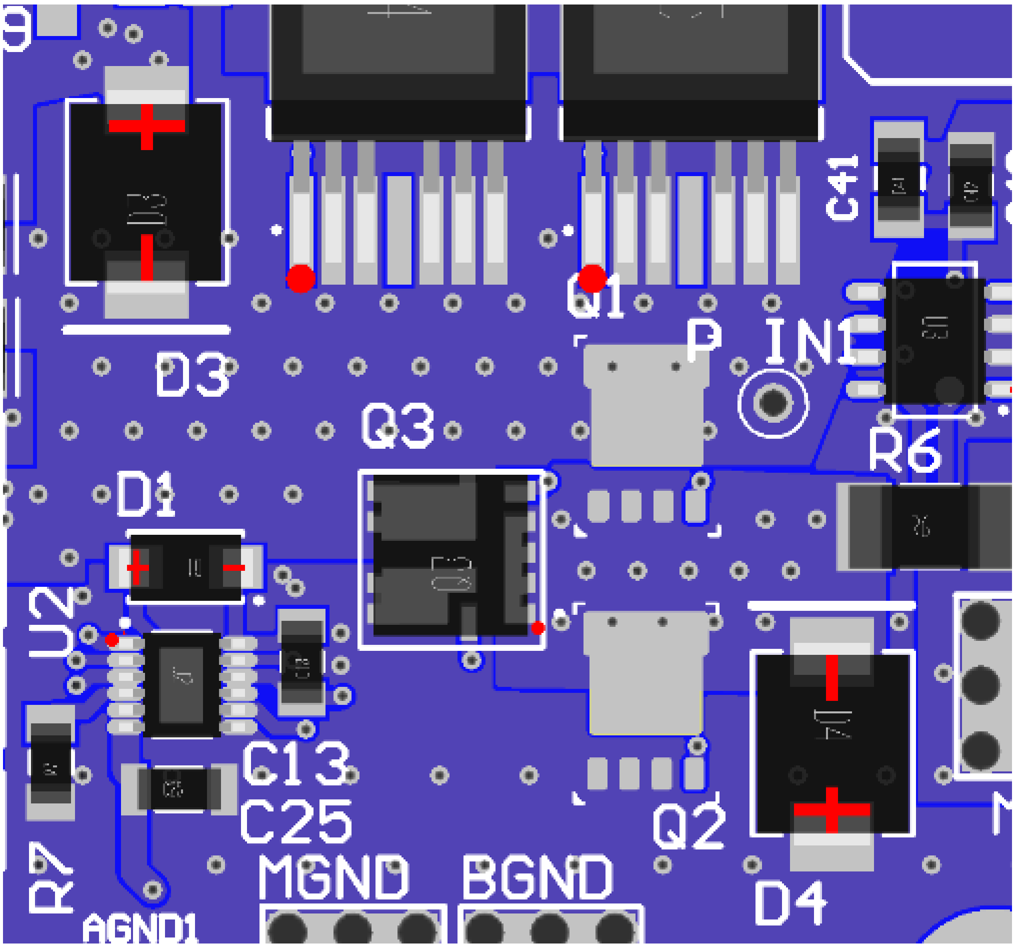


Figure 5.1-1 Confirmation message from JLCPCB on diode orientation.

Desoldering and resoldering the diodes in the correct orientation fixed the issue. All the other components worked first time on the power electronics board.

5.1.2 Layout

This issue was spotted soon after the PCB entered production but too late to cancel the order. The +11V power rail is shorted to the isolated ground test point, this was a relatively simple fix cutting the trace on either side and using enamel wiring to shunt across. In principle this is likely not an issue but it still breaks isolation and is not ideal additionally the specific working principle of the isolated DCDC might cause an issue with the isolated ground having the same potential as the supply voltage.

5.1.3 Operational Amplifier

The operational amplifier used on the power controller board is an 8 pin 2 amplifier design, this design has the u741 footprint layout. As can be seen from figure 5.1-2 the power supply pins were connected the wrong way around with 5V connected to the V- pin and GND to V+.

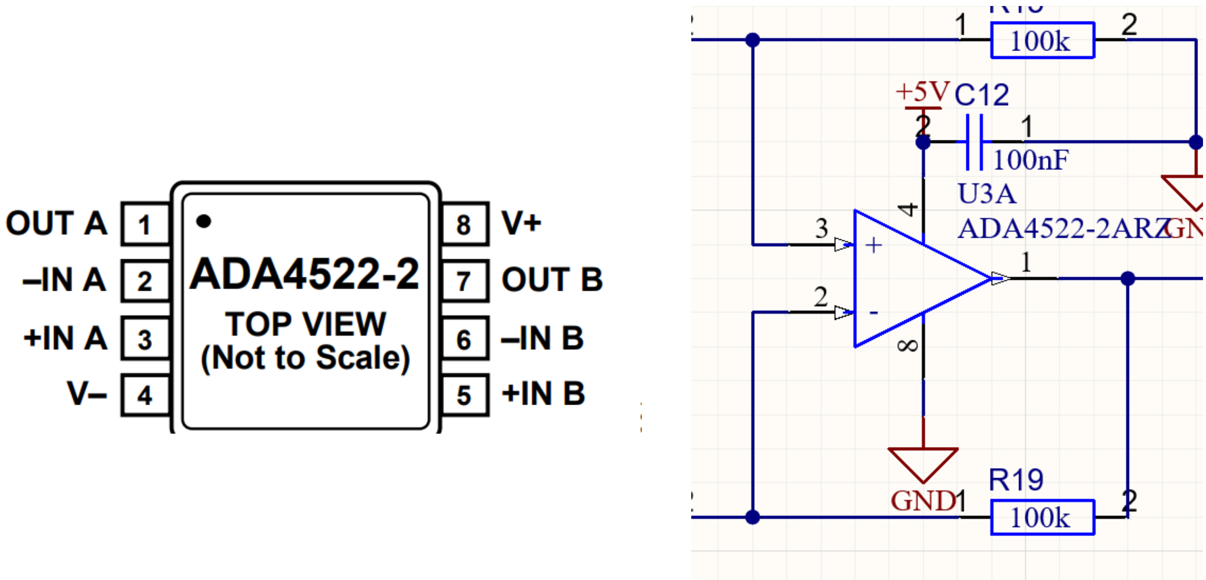


Figure 5.1-2 IC pinout and original schematic of U3 Op amp

Performing a wire modification as seen in figure 5.1-3 was required to test the circuit. After multiple attempts and separate PCBs, the circuit was verified to work. The IC would immediately fry if the pins touched the wrong connection. This happened after verifying the circuit working. Once the circuit was verified, a new revision was ordered with the fix implemented.

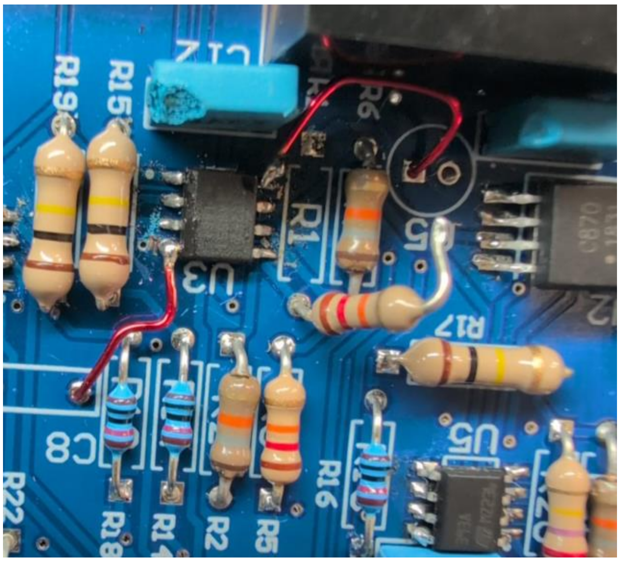


Figure 5.1-3 Wire modification of the original power controller board

5.1.4 Hand Assembly

Initial experiments with the working PCB had issues surrounding the I^2C DAC, this led to two pieces of information to learn.

1. The PC PCB lacked test points which could've helped debug the circuit.

2. The tiny footprint of the DAC was difficult to solder accurately. Each pin is roughly 1mm wide. Using a USB microscope it's evident that some pins are not fully soldered. Due to issues around bridging and this is a difficult task.

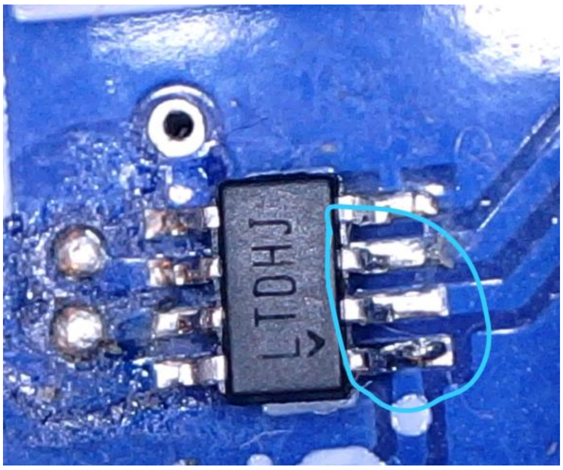


Figure 5.1-4 Cold solder joint on SOT-23-8 footprint.

Tapping both side of the IC with the soldering iron to reflow the solder. Fixing the communication issues.

5.1.5 Logic Level

The D-type Flip flop (SN74AC74) is a standard logic gate with an input supply range from 2-6V. However, on a 5V supply V_{IH} is ~3.7V. The ESP32 is a 3.3V microcontroller, despite this the flip flop worked perfectly for a week or two of testing. This stopped working shortly before the final track test. The solution was relatively simple, a standard bidirectional logic translator.

Using the exposed GPIO on the power controller board the circuit in figure 5.1-5 was also implemented. This restored functionality to the flip flop. In a redesign or rebuild of the control board, the SN74ACT74 would be used instead as this flip flop has fixed logic levels independent of supply voltage.

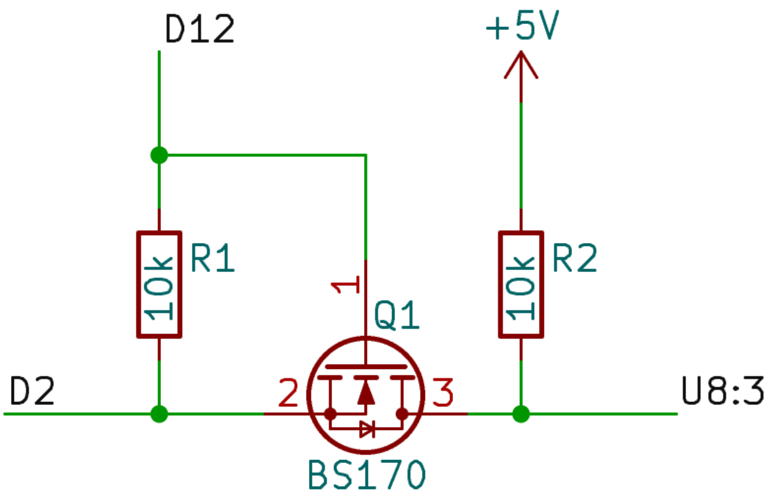


Figure 5.1-5 Logic translator implemented on the power controller.

5.1.6 Low Pass Filtering

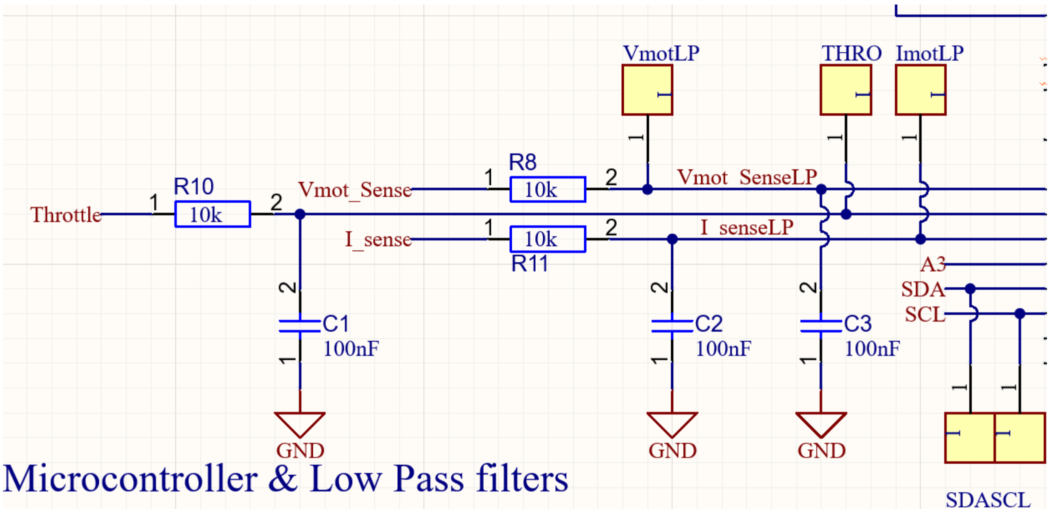


Figure 5.1-6 Low Pass filter on the Analog signals.

When the low pass filter is connected to the circuit with the Arduino nano ESP32 there is a voltage on the node pin almost like a voltage divider between the internal signal. The resistor R8, R10 R11 had to be reduced to 510Ω. The effective cutoff frequency is 3kHz, arguably a little bit low for the high frequency elements, Figure 5.1-7 shows the ripple current apparent in the waveform. This is particularly an issue as the samples are only taken every 100ms.

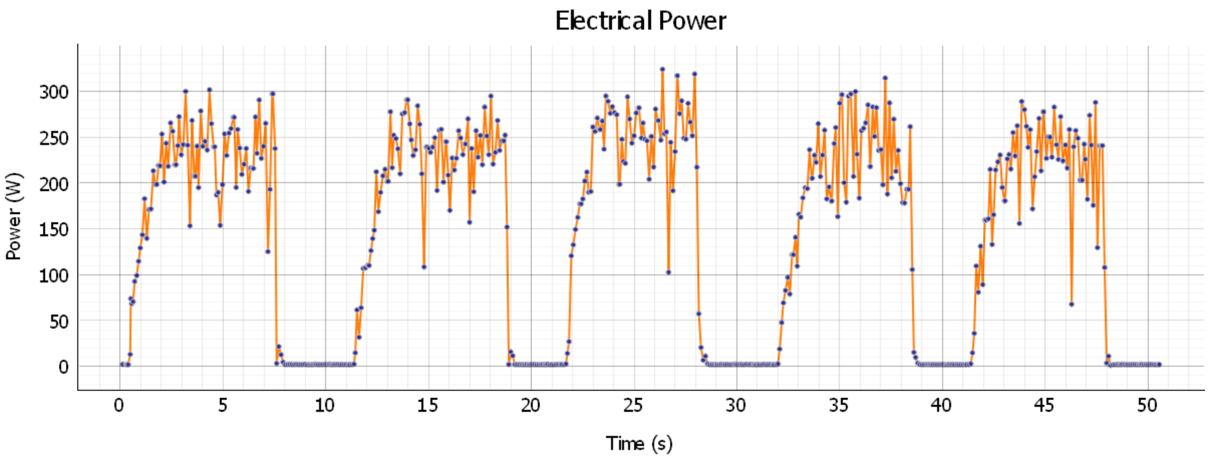


Figure 5.1-7 Noise on the analogue current input.

5.2 Physical Testing

Once the issue identified in section 5.1.1 was resolved the LEDs could be tested as part of the initial bring up. The LEDs are only active when the shunt is placed across the two jumper pins.

DS1 should have the PWM label underneath instead of SOFT. The LED clearly gets brighter and dimmer depending on the duty cycle. At 30kHz the minimum duty cycle was 0.03%.

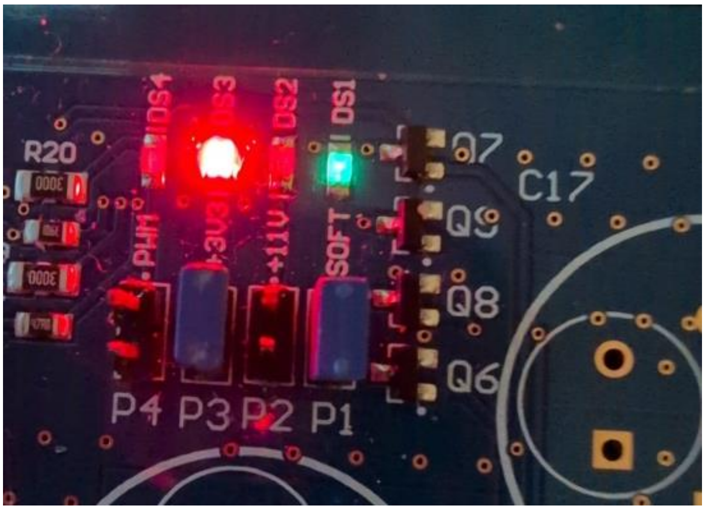


Figure 5.2-1 LED bring up.

All circuits were tested and each area was verified first with a multimeter and then an oscilloscope. Before attaching a significant load, the output was tested on a high-power resistor.

5.2.1 Power Consumption

Previous Power Electronics		
Set Up	Power(W)	Energy(J)
Power Electronics	0.91	2,185
Power Electronic + DAQ	2.56	6,150
Battery Input (incl. relays)	4.68	11,248

New Power Electronics		
Set Up	Power(W)	Energy(J)
Power Electronics Board	0.225	540
Power Electronics + 2mA/LED	0.425	6,150
PE + PC Boards	0.975	2,340

New DAQ has not been measure combined with the PE and PC.

5.2.2 Current Control

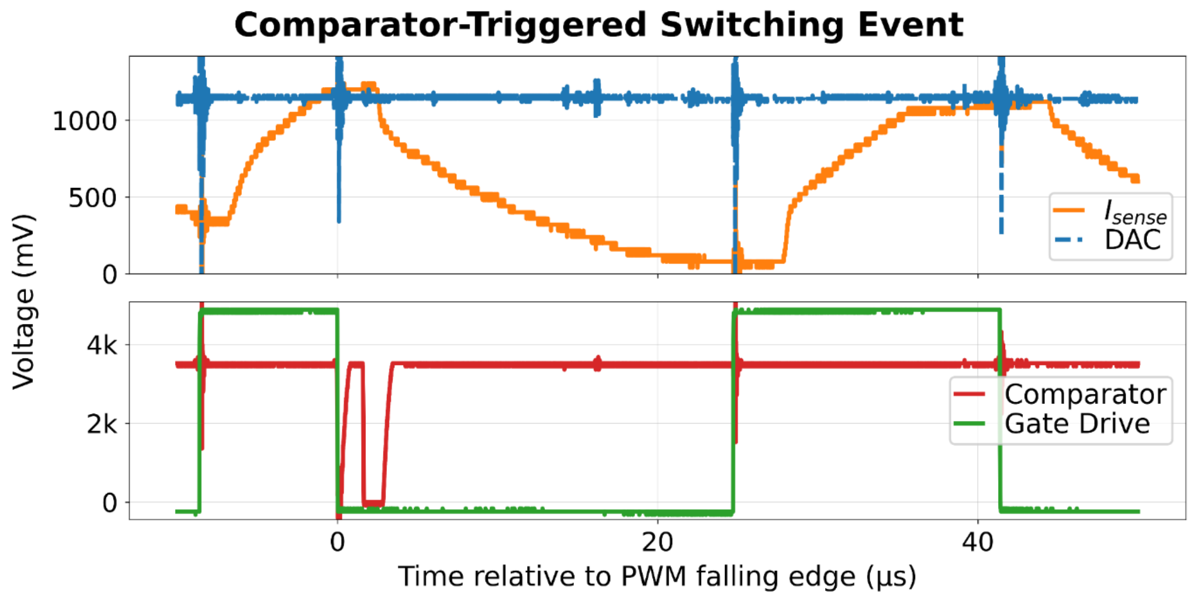


Figure 5.2-2 Switch node cutoff and latch resistance.

Figure 5.2-2 shows the current control for a target duty cycle of 50% with a load acting against the motor on the dyno. At 0s the comparator pulls low, due to inductive ringing on the DAC voltage the comparator briefly returns high but due to the latch on the flip flop the output gate drive signal is unaffected.

The full 50% waveform runs without the comparator triggering. A key visible issue is the ringing on the switching events in the DAC line but that can also be seen in the comparator line.

5.2.3 Combined Dynamometer Test

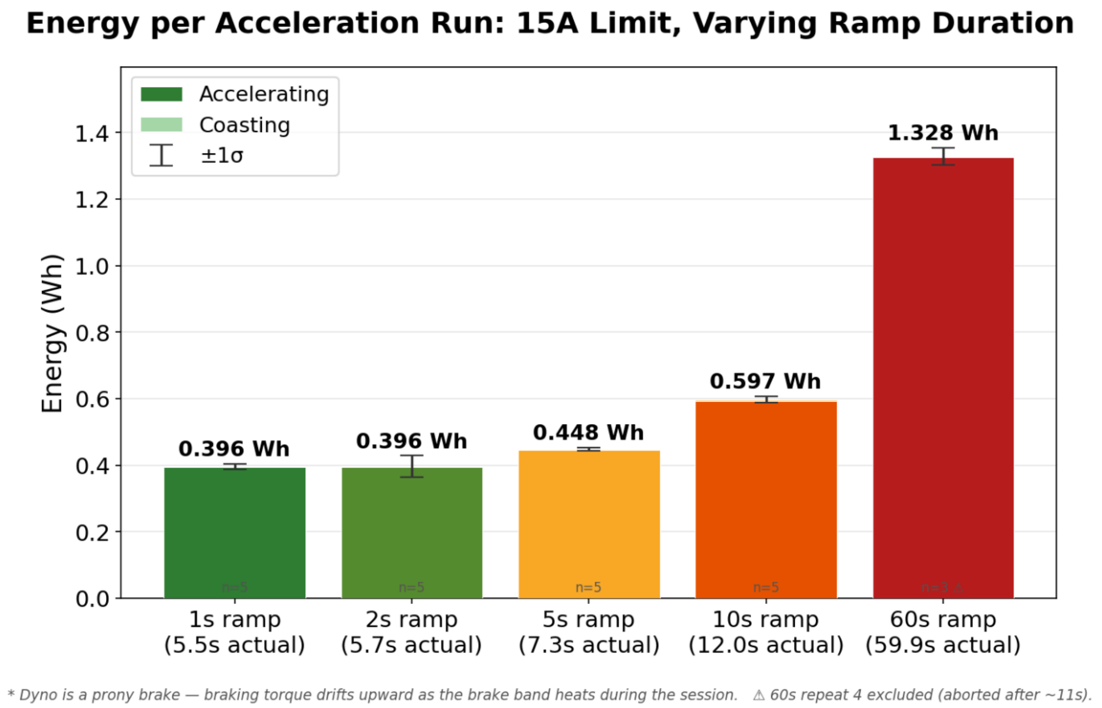
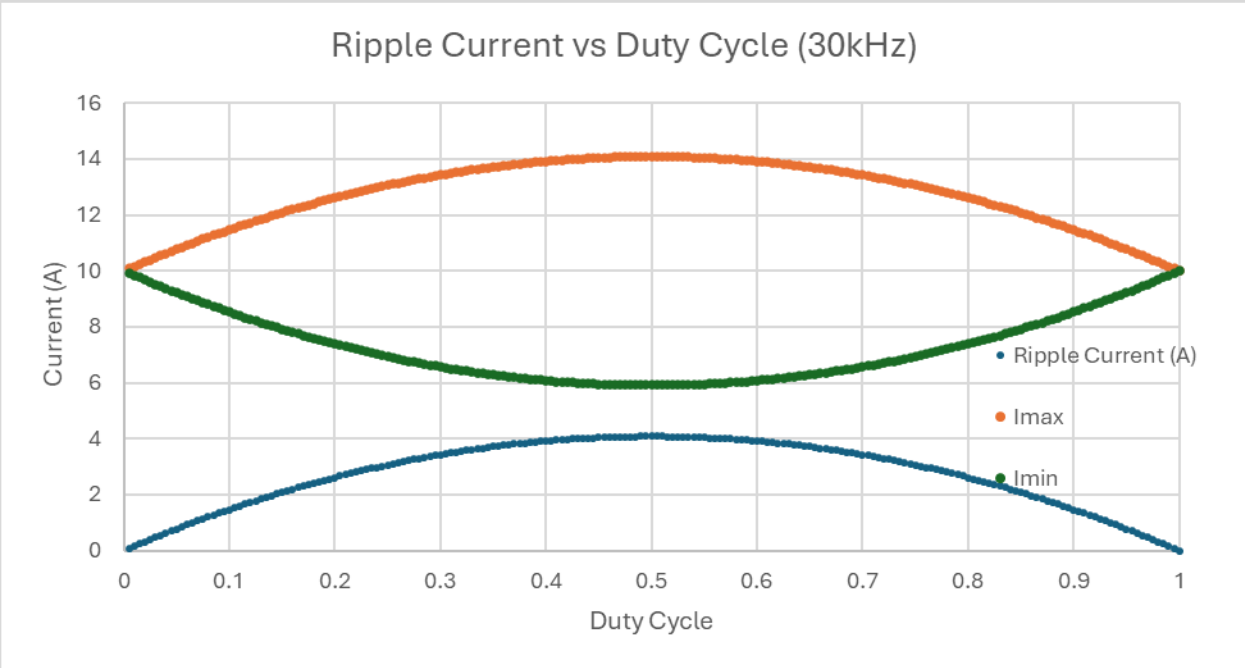


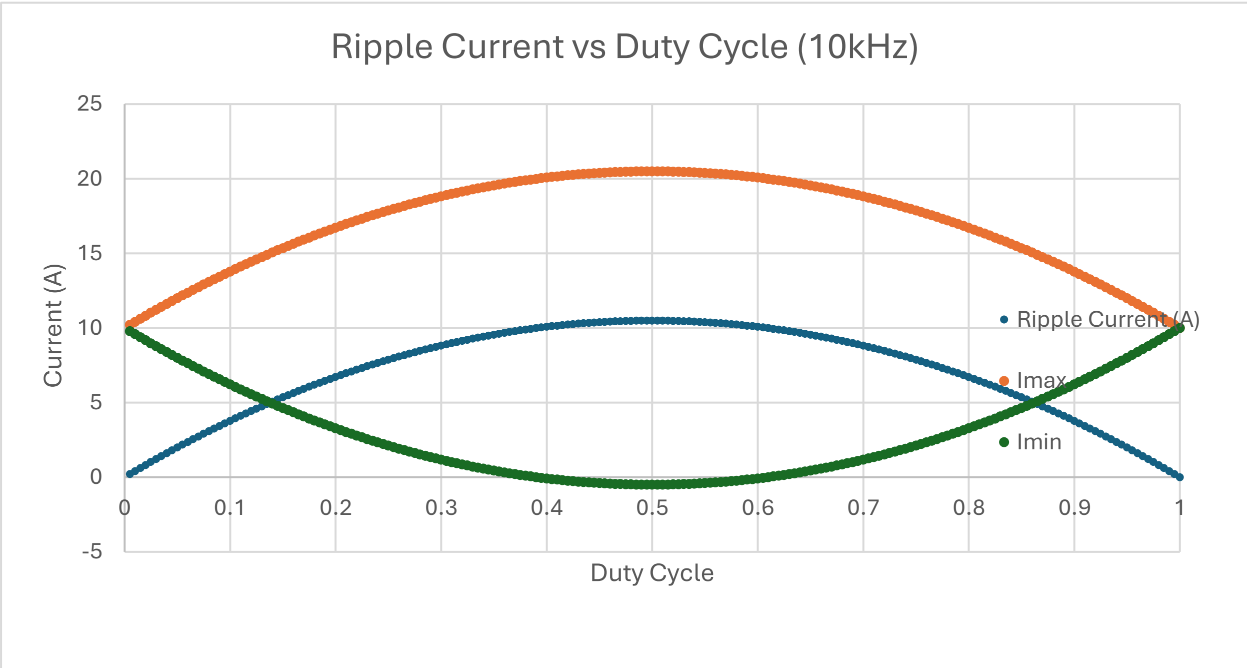
Figure 5.2-3 Energy consumed for the fixed current limit and emulating the open loop ramp rate of the previous power electronics.

The results above show that for a fixed current limit on a constant torque load acting against the motor. There are still energy improvements to be had in comparison to the open loop setup as expected before.

Even in comparison to the race conditions of the 5s ramp rate there's still a 12% improvement in energy consumption. Comparing this to the dynamic conditions expected on a race track, this is bound to show further energy improvements.

5.2.4 Switching Frequency





The graphs above show the expected ripple current on through the shunt resistor based on the duty cycle, switching frequency and terminal inductance. In the case of the negative current for the 10kHz this would manifest as 0A rather than negative current flow.

Figure 5.2-2 shows the oscilloscope capture of the current at the switch node. The yellow line is the high bandwidth instrumentation amplifier. In the left graph the signal corresponds to 20A peak to peak. The right graph shows 8A peak to peak with a DC offset of 5A minimum. Using the ESR of the motor terminals (0.103Ω) this corresponds to 11W vs 8.5W between the 10kHz and 45kHz.

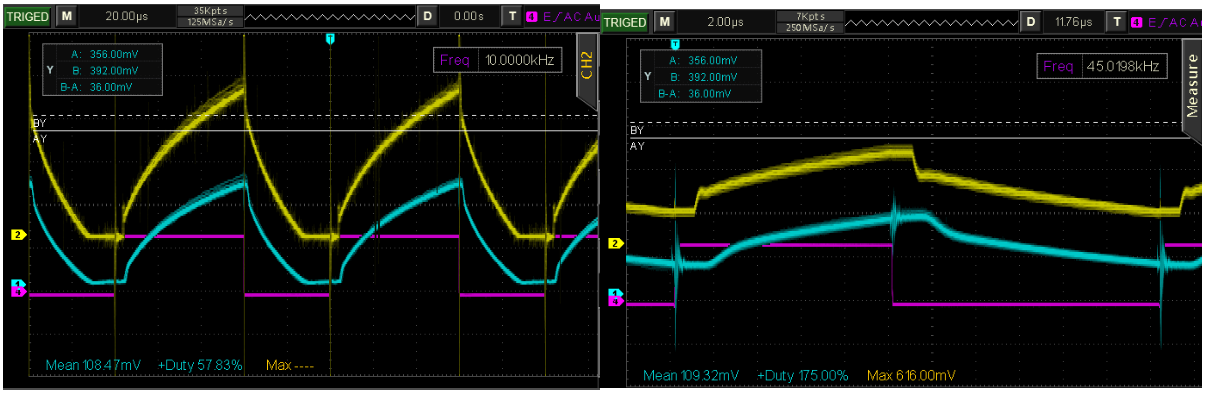


Figure 5.2-4 Oscilloscope capture showing the ripple current at roughly 50% duty cycle.

Running the experiment code as discussed in section 4.5.2, Figure 5.2-4 shows the energy consumption at different switching frequencies for the same load and at two different current limits. The difference in energy consumption is marginal between the different current limits. I do think the issues surrounding the dyno speed update frequency is a partial issue and could explain some strange results.

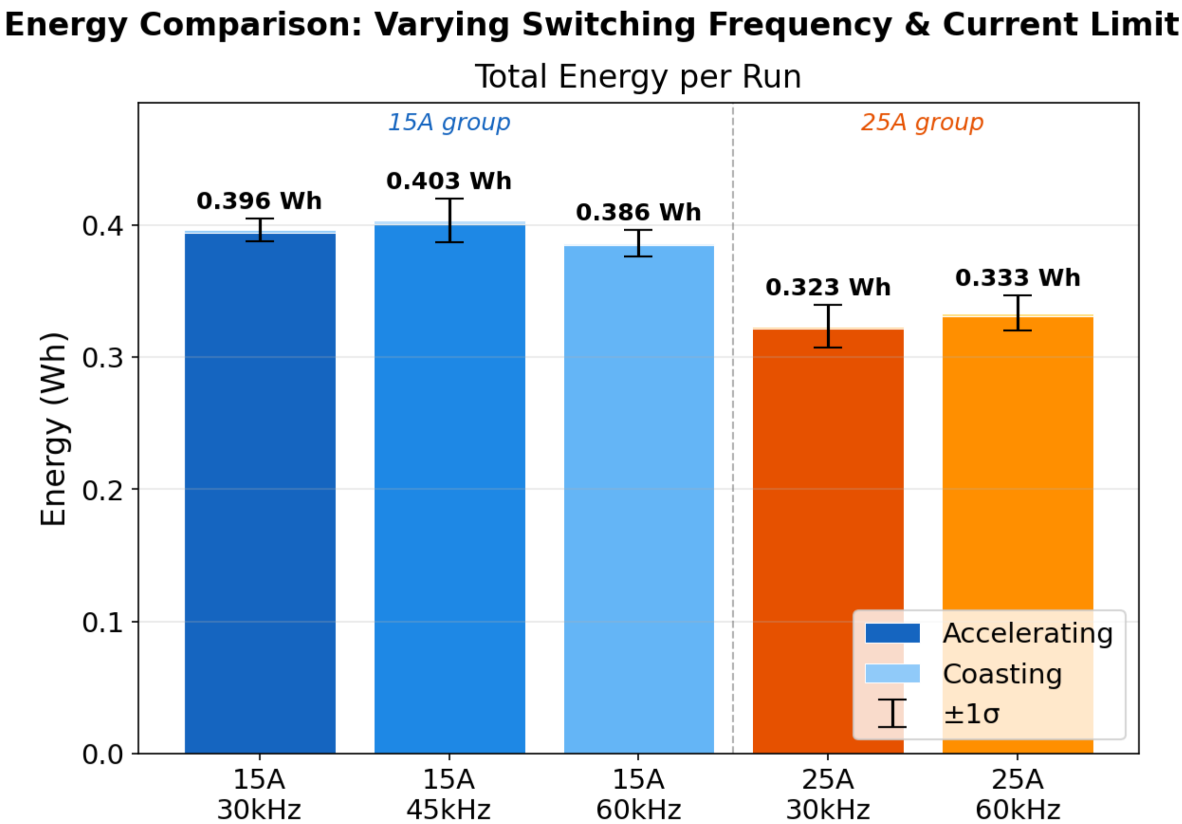


Figure 5.2-5 Energy Output for current limit at different switching frequencies.

6 Conclusion

The new power electronics

6.1 Recommendation For Future Work

Within individual sections I've already detailed some additional work that can be completed. Other areas that could be tested.

- **No Microcontroller:** This circuit [ADI] [TI] can be used to create a PWM signal without any microcontroller. This would remove any need for a microcontroller and remove a large current sink on the PCB
- **µC Sleep:** Adding a sleep mode to the ESP32 when the throttle hasn't been used in a second and entering a low power mode would be useful and remove some constant energy consumed by the Geec.
- **Dyno Speed:** Changing the method for calculating speed from the dyno and the DAQ can improve response rate. The method is a telescoping method. Instead of capturing the number of pulses per second or the time between pulses, the time stamp of each pulse is recorded to a circular buffer. Taking the difference between the first and last timestamp will get the mean sample divided by the length of the buffer. The main benefit of this is that the signal is much less quantised, using micros and allows new values to be calculated up to the update frequency of the buffer.
- **Acceleration Modes:** Using the button on the steering wheel the driver should be able to communicate with the PE to set an acceleration mode between aggressive, conservative etc. This could be most useful for initial acceleration which needs more power and torque than the regular accelerations around the track.
- **GaNFETs:** The graph's above show similar to better performance at higher switching frequencies, this could further incentivise the transition to GaN HEMTs. This setup would require automated assembly due to the size of most GaNFETs.

6.2 Acknowledgements

I would like to thank Prof Maeve Duffy for providing technical support and guidance. Myles Meehan and Darragh Mullins for their constant availability and support when different problems arose (usually all at once). Dr Nathan Quinlan and Dr Martin Glavin for supporting the Geec additionally. The other Geec thesis and project leads but particularly Molly Hennessy for supporting me managing the team. The wider Geec team for putting in massive amounts of work for the enjoyment of the project without any academic benefit.

I'd like to thank my family for supporting me through college and setting me up to learn and engage as I have. I'd also like to thank my friends who've helped me all the way through college, with assignments, hobbies and activities.

Thank you all.

7 Appendix

Here is a link to the github repo that has also been copied to the Geec Onedrive.

https://github.com/cononinator/GeecPE2526

7.1 Use of AI

I have used AI in this project in a few areas. The first is to accelerate the programming, especially the Python experiment code. Some initial investigations began with the use of AI e.g. determining a method for current limiting, this was then moved to verify and testing in simulation before moving to schematic design and layout.